ABSTRACT

The Travel Insurance Management System is a full-stack web application developed using Angular, Tailwind CSS, and ASP.NET Core 8 Web API to digitize and manage travel insurance policies and claims efficiently. The system is designed with a secure role-based access control mechanism supporting four roles: Admin, Customer, Agent, and Claims Officer.

Users who register through the application are assigned the Customer role by default. A single Admin account is seeded during system initialization and has full authority to manage policies, configure pricing logic, and control user roles. The Admin can promote registered customers to either Agent or Claims Officer roles based on organizational requirements, ensuring controlled privilege management and preventing unauthorized role escalation.

Customers can browse available travel insurance policies such as Single Trip, Multi-Trip Annual, Family, and Student plans. The system dynamically calculates premiums based on factors including age, travel duration, and destination risk. Upon successful payment, policies are automatically generated to maintain data consistency and transactional integrity. Customers can also submit claims and track their status online.

Agents function as sales representatives who assist customers and monitor policies associated with them, while Claims Officers are responsible for reviewing submitted claims, verifying supporting documents, and approving or rejecting claims according to defined coverage rules.

The backend is implemented using ASP.NET Core 8 Web API with JWT-based authentication and role-based authorization to ensure secure access control. The frontend, built using Angular and Tailwind CSS, provides a responsive and user-friendly interface. The system demonstrates practical implementation of full-stack development, dynamic premium calculation, secure role management, and structured workflow design within the travel insurance domain.